



Proliferated Warfighter Space Architecture
Tranche 3 Transport Layer – Upsilon (T3TLv)
DRAFT Program Solicitation
22 January 2025

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1 Introduction

The Space Development Agency (SDA) is issuing this solicitation to establish the foundation for Tranche 3 (T3) of the Proliferated Warfighter Space Architecture (PWSA). The PWSA's operational utility is predicated on the availability of a ubiquitous data and communications Transport Layer provided by a proliferated constellation of relatively small, mass-producible Space Vehicles (SVs) in Low Earth Orbit (LEO). The Tranche 3 Transport Layer (T3TL) SVs will be similar to those currently under development for the Tranche 2 Transport Layer (T2TL), with targeted technology enhancements, mission-focused payload configurations, increased integration, and greater production efficiencies. T3TL will provide global communications access and deliver persistent regional encrypted connectivity in support of Warfighter missions around the globe.

Specifically, T3TL features multiple SV and mission configuration variants procured through a multi-solicitation and multi-vendor acquisition approach. This solicitation is for the Upsilon (υ) variant of the T3TL as summarized in Table 1-1.

Table 1-1 – T3TL-υ Constellation

T3TL Baseline Mission Payloads and Subsystems (included on all T3TL SVs)			
• (U) Four (4) Optical Communication Terminals (OCTs) • (U) Multi-user Ka-band mission payload • (U) Networking and data routing subsystem • (U) Navigation subsystem • (U) S-band backup Telemetry, Tracking and Control (TT&C) subsystem			
T3TL Variant	Unique Mission Payloads and Capabilities	Total SVs	Anticipated Number of Awards/Vendors
(U) Upsilon (υ)	• (U) S-band TACSATCOM mission payload • (U) UHF TACSATCOM mission payload • (U) Integrated Broadcast Service – LEO (IBS- L) mission payload • (U) Global Navigation Satellite Systems (GNSS) Situational Awareness (SA) mission payload	(U) 40	(U) Two (2); 20 SVs for each Vendor

1.1 Motivation

SDA has determined that the proliferation of various mission functions across the PWSA using commodity SVs is the most efficient and cost-effective means of evolving the Architecture. To accomplish this, SDA employs a capability-focused business model prioritizing speed and lowering costs by harnessing commercial development to achieve proliferation and enhance resilience. The objective is to establish and demonstrate the ability to rapidly evolve development processes and field capability on a significantly faster timeline than historical space systems acquisition. In pursuit of this objective, SDA requires each PWSA Performer's SVs and communications systems to be interoperable with the SVs and systems developed by all other Tranche 3 PWSA performers as well as those to be deployed in Tranche 1 and 2. Additionally, all SVs must operate in an integrated fashion through a common ground architecture.

1.2 Program Approach

This solicitation seeks Other Transaction (OT) proposals under the authority of 10 U.S.C. § 4022 for U.S. space industry builder-operator teams that will partner with SDA to jointly develop and operate the T3TL_v SVs and mission as part of the PWSA. The Government intends to purchase a constellation of 40 T3TL_v SVs, divided into four (4) orbital planes of ten (10) SVs each, to be awarded to two (2) Offerors, subject to available funding. Each Offeror shall propose to develop two (2) orbital planes of ten (10) T3TL_v SVs per plane, along with the associated ground systems and operations and sustainment capability.

Note that for SDA to make an OT award to an Offeror, at least one of the following conditions must be met:

- (A) There is at least one (1) nontraditional defense contractor or nonprofit research institution participating to a significant extent in the prototype project.
- (B) All significant participants in the transaction other than the Federal Government are small businesses (including small businesses participating in a program described under section 9 of the Small Business Act (15 U.S.C. 638)) or nontraditional defense contractors.
- (C) At least one third of the total cost of the prototype project is to be paid out of funds provided by sources other than the Federal Government.
- (D) The senior procurement executive for the agency determines in writing that exceptional circumstances justify the use of a transaction that provides for innovative business arrangements or structures that would not be feasible or appropriate under a contract, or would provide an opportunity to expand the defense supply base in a manner that would not be practical or feasible under a contract.

For more information on Other Transactions please see: <https://www.ecfr.gov/current/title-32/part-3>

1.3 Program Plan and Schedule

The first two planes of the T3TL_v constellation will be launched no later than (NLT) September 1, 2028. The Offeror is expected to plan their program in accordance with the milestone dates and deliverables specified in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document of this solicitation.

1.4 T3TL Concept of Operations (CONOPS) Overview

The T3TL establishes Initial Launch Capability (ILC) in September 2028 with the launch of two planes of T3TL_v SVs and continues with an approximately year-long, monthly launch campaign involving the multiple T3 variants. As these SVs are placed into their insertion orbits, SDA will undergo a continuous checkout and commissioning process to prepare Tranche 3 for acceptance into operations conducted out of the Operations Centers (OCs) at Grand Forks, AFB (OC-N) and Redstone Arsenal, AL (OC-S). These OCs will be operational in time to support Tranche 1 – the PWSA’s initial warfighting capability for Beyond Line of Sight (BLOS) targeting and data transport and advanced missile detection and tracking. Tranche 1 and Tranche 2 systems will be conducting mission operations in the OCs at the time that Tranche 3 achieves ILC.

After being placed into an insertion orbit by Government procured NSSL launch vehicle, each plane of SVs will undergo Launch and Early Operations (LEOPS), which includes the Separation /

Initialization / Checkout / Orbit Raising (SICO) phase and the Functional Acceptance Test (FAT).

SICO includes initial acquisition of the SVs, SV and payload initialization and aliveness checks, orbit raising and phasing to the operational orbit, and Offeror preparation for Government, and Tranche 3 Program Integrator-led acceptance testing.

FAT is then conducted to ensure that each system delivered by the Offeror – comprised of a plane of SVs integrated with the Offeror's Network Established Beyond the Upper Limits of the Atmosphere (NEBULA) Operations – Vendor Architecture (NOVA) ground system – is fully mission capable. Once FAT and the subsequent Functional Acceptance Review (FAR) are successfully completed, each plane of Tranche 3 SVs will be transitioned to the Operations phase, with operations conducted from the OCs.

T3 commissioning activities will be conducted by the Offeror in coordination with the Tranche 3 Program Integrator, and these activities will be conducted from the Offeror's facilities in T3.

SUPERNOVA instances serving LEOPS may exercise a subset of the operational SUPERNOVA's functional capabilities but will replicate critical elements of the SUPERNOVA-NOVA interface and many key capabilities required to assess NOVA and SV functional capability prior to transition to Operations.

SDA envisions the "exit state" of the commissioning phase to be very similar to the "entry state" to Operations – a fully-functional, optically-interconnected, and networked plane of SVs commanded and controlled by Offeror personnel using their NOVA ground system with verified interfaces and connectivity with SUPERNOVA.

Once commissioning is complete, each plane of SVs will undergo a similar set of functional tests from the OCs during a brief Operational Acceptance Test (OAT) and begin their transition to Mission Operations. The Offeror will continue to support their vehicles for a period of up to five (5) years following the transition to Mission Operations.

2 System Description

The T3TL_U constellation will consist of four (4) orbital planes with ten (10) T3TL_U SVs in each plane. The constellation parameters are defined in Table 2-1. Every T3TL_U SV will be interoperable with all PWSA SVs, regardless of vendor. The T3TL_U SVs will include Optical Communications Terminals (OCTs) to support in-plane and cross-plane crosslinks, links to terrestrial OCTs (ground, air, and maritime), and crosslinks to compatible SVs external to the PWSA. The T3TL_U SVs, in conjunction with the SDA OCs and the T1, T2, and other T3 SVs, will form a communications network that will provide resilient, low-latency, high throughput data transfer to and from any location on the globe.

Table 2-1 – Nominal T3TL-Upsilon Constellation

Parameter	Value
Number of Planes	4
SVs per Plane	10
Altitude	1020 km
Inclination	~45 deg
Nodal separation between planes	~45 deg
In-Plane Mean Anomaly Separation	36 deg
Anomaly Phasing Between Consecutive Planes	~4.5 deg

The T3TL_v SVs will also have the capability to provide advanced communication and data transfer via S-Band TACSATCOM, UHF TACSATCOM, and IBS-L links. This includes the ability to relay messages beyond-line-of-sight between the SDA OC and in-theater ground, maritime, and airborne users.

The system description and requirements can be found in Attachment 3 – Technical Requirements Document (TRD) of this solicitation. The Offeror shall propose a solution that meets the requirements as laid out in that attachment.

SDA expects Offerors to propose an insertion orbit for Plane 1 with a semi-major axis between 6898 km and 7228 km. The expected mass capacity of the launch vehicle to each of the insertion orbits is specified as a function of the semi-major axis of the Plane 1 insertion orbit in Table 2-2. For Plane 1 insertion orbit semi-major axis values not specified in Table 2-2, the Offeror may assume mass capacity values that result from linear interpolation. The mass capacity requirements in Attachment 3 – TRD will be updated in accordance with values computed from agreed upon approved Plane 1 insertion orbit parameters and calculation of the associated mass capacity values.

The Offeror-proposed insertion and orbit raising approach is expected to enable the SVs to comply with the requirements for time to operational orbits from Attachment 3 – TRD.

Table 2-2 – Insertion Plane vs Mass Trade

Parameter	Insertion Plane 1 Semi-major axis (km)	Insertion Plane 2 Semi-major axis (km)	Max Mass to Plane 1 Insertion Orbit (kg) 20 SVs + leave-behind mass	Max Mass from Plane 1 Insertion Orbit to Plane 2 Insertion Orbit (kg) 10 SVs + leave-behind mass
Baseline	6,898	7,348	11,100	6,100
Option 1	7,078	7,348	11,000	6,000
Option 2	7,128	7,348	10,900	6,000
Option 3	7,178	7,348	10,800	5,900
Option 4	7,228	7,348	10,500	5,800

3 Selection Process

Within this solicitation, the following definitions are used:

- **Prime:** The Offeror
- **Major Teammate:** A member of the Prime's development team with a different CAGE code from the prime Offeror, that either accounts for no less than five (5) percent of the proposed cost and/or develops one or more of the following subsystems or capabilities:
 - Optical Communications Terminals
 - IBS-L payload (if separate from TACSATCOM)
 - S-band TACSATCOM payload (if separate)
 - UHF TACSATCOM payload (if separate)
 - Space Vehicle Bus
 - System Integration
 - Operations and Sustainment Personnel
 - NEBULA Operations – Vendor Architecture (NOVA) System

If the following subsystems are not procured as commercial off-the-shelf systems (COTS) systems and/or require significant modifications, then the developer of the subsystem shall be considered a Major Teammate:

- Networking and Data Routing Subsystem
- Encryption Subsystem
- Ka-band Mission Payload
- GNSS SA Mission Payload

Major Teammate proposals (regardless of teammate tier) shall contain the information requested in Section 3.1.4.4 to allow the Government to review the proposed price and supporting rationale. Major Teammates shall not submit Governance or Technical (or other) volumes. Note that certain information regarding Major Teammates is requested elsewhere in the solicitation, but this information should be included as part of the Prime's proposal.

Major Teammate proposals shall follow the same font, spacing, table, and margin formatting instructions that apply to the Prime.

- **Commercial off-the-shelf:** COTS products are defined as software or hardware products that are commercially ready-made and available for sale, lease, or license to the general public. For the purposes of this solicitation, products that are less than TRL 7 cannot be considered COTS. If a product requires substantial modifications as part of the T3TL_v program, then it cannot be considered COTS.

4 Proposal Preparation Instructions

The Offeror shall submit their T3TL₀ proposal via six (6) volumes. Specific volume contents and associated page limits are detailed below:

Volume/Tab	Title	Page Limit	Due at 1200 EST on date specified below
Volume I	Governance	20 pages	28 April 2025
	Table of Contents/Cross Reference Matrix		
Tab 1	Executive Summary		
Tab 2	Program Management		
Tab 3	Security Performance		
Tab 4	Facilities Management		
Tab 5	Systems Engineering		
Tab 6	Mission Assurance		
Tab 7	Organizational Conflict of Interest (OCI)	Excluded from page count	
Tab 7	Key Personnel Résumés	Excluded from page count	
Tab 8	Personnel Matrix		
Tab 9	Technical Data and Computer Software Rights Assertion	Excluded from page count	
Volume II	Technical	35 Pages (Excludes Appendices)	28 April 2025
	Table of Contents/Matrix		
Tab 1	Executive Summary		
Tab 2	CONOPS Overview		
Tab 3	Space Vehicle Overview		
	• T3TL ₀ Payload Architect. & Design		
	• IBS-L Mission Payload		
	• TACSATCOM Mission Payloads		
	• Optical Communications Terminal (OCT)		
	• Ka-band Mission Payload System		
	• Telemetry, Tracking and Control (TT&C)		
	• Network and Data Routing		
	• Red-Side Processing (RSP)		
	• GNSS SA Mission Payload		
Tab 4	Space Vehicle Bus Design		
	• Bus Design and Heritage		
	• Space Vehicles (SV) Subsystems		

	• Cybersecurity		
	• Encryption		
	• Navigation		
Tab 5	Manufacturability		
	• Supply chain		
Tab 6	Test, Demonstration and Verification		
Tab 7	Launch Services Overview		
Tab 8	Ground Segment Integration and Operations		
Tab 9	Mission Operations, Sustainment & Disposal		
Tab 10	Appendices		
	• TACSATCOM Payload Design Tech Vol Response	5 Pages	
	• Multi-Carrier IBS-L Mission Payload Design	3 pages	
	• State of Charge Analysis	5 pages	
	• Mission Availability and Reliability	5 pages	
	• IBS-L and TACSATCOM Mission Payload RF Link Budgets	Unlimited	
	• Launch, Early Operations, Commissioning, and Transition to Operations Plan	5 pages	
	• Commercial Data Sheets	20 pages	
	• Parameter Specifications	Excel	
	• Technical Requirements Verification Methods		
Volume III	Schedule	10 Pages (Excludes Appendix)	28 April 2025
Tab 1	IMS		
Tab 2	Schedule Risk		
Tab 3	Appendix – Portfolio IMS	5 pages	
Volume IV	Price and Rationale	Unlimited	28 April 2025
Tab 1	Executive Summary and Pricing Rationale		
Tab 2	Price Proposal Structure		
	• Bidding Unit		
	• Work Breakdown Structure (WBS)		
	• Milestone Payments		
	• Delivery Incentive		
	• On Orbit Penalty Performance		
Tab 3	Proposal Pricing Details		
Tab 4	Govt Furnished Equipment, Information & Property		
Tab 5	Major Teammate Proposals		
Tab 6	Organizational Conflict of Interest (OCI)		
Volume V	Past Performance		28 April 2025

Tab 1	Executive Summary	1 Page	
Tab 2	Recent and Relevant Past Performance		
	Citations (up to 4) to include POC info	4 pages each (NTE 16 pages)	
Volume VI	Draft Other Transaction Agreement (OTA)	Unlimited	28 April 2025
Note: Page numbering shall re-start at the beginning of each section or volume. Note: Classified Addendums are included in the page count identified above. Note: Cover pages, tables of contents, appendices and acronym tables are not part of the page count in any volume and are optional for Volumes III-VI			

The Offeror should focus on demonstrating the depth of their design and demonstrating through analysis that it achieves the necessary performance, rather than echoing all Government requirements in their response.

- **Volume I – Governance** shall contain a detailed description of the Offeror’s proposed program management, systems and mission engineering approaches, processes, and tools for executing the program on schedule and within the price proposed. This volume shall also contain résumés for Key Personnel that will perform the T3TLv program and a complete listing of data rights asserted by the Offeror and all its subcontractors.
- **Volume II – Technical** shall contain a detailed description of the Offeror’s proposed technical solution to accomplish the requirements defined in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document.
- **Volume III – Schedule** shall contain a complete Integrated Master Schedule. This volume shall also provide an analysis of schedule risks, identification of key milestones, entrance criteria, key accomplishments, and exit criteria for significant reviews and activities, and a clear presentation and analysis of the critical path(s) from contract award through SV delivery.
- **Volume IV – Price and Rationale** shall contain the Offeror’s Firm Fixed Price and associated detail.
- **Volume V – Past Performance** should contain at least one (1) and may contain up to four (4) recent (within the last five (5) years) relevant examples of on-time, on-cost/price delivery of complex systems, subsystems, or components, preferably space-related hardware or software. Offerors should also include at least one (1) and up to two (2) Past Performance examples for each Major Teammate.
- **Volume VI – Draft Other Transaction (OT) Agreement**

Normal text in all volumes shall be in Times New Roman font, and the font size shall be no smaller than 12 pt. All text in figures and tables for all volumes shall be in Arial Font, and the font size shall be no smaller than 8 pt. A table shall consist of at least two (2) columns in every row. Tables or figures that are deemed attempts to circumvent these text formatting restrictions will be removed from the proposal and not evaluated. All margins (top, bottom, left, and right) shall be no smaller than one (1) inch. Single-line spacing shall be used for all text in paragraphs, bulleted lists, and tables. All .pdf pages, in all volumes, formatted larger than 8.5 inches by 11 inches shall be counted as two (2) pages. Line numbers shall be applied to all text sections in all volumes and shall be set within the left margin of the page so as not to reduce the usable size of the page. Title pages and tables of contents, tables

of figures, and tables of tables will not count against the page counts of the respective volumes and will not be evaluated for content. **Note: Nonconforming proposals may be rejected without review.**

- A Table of Contents shall be incorporated into Volume I & II and will include a listing of the section titles and subsection titles. Page numbers for each title must be indicated. The Table of Contents shall also include a listing of illustrations and tables with page numbers.
- A cross-reference matrix for Volume I & II is also required. This matrix shall indicate the proposal volume section, subsection, and paragraph, cross-referenced with the appropriate SOW paragraphs, applicable solicitation instructions; evaluation criteria, and deliverables/CDRLs, as applicable.
- A glossary of acronyms and terms for Volume I & II is required. The use of acronyms without definitions within the proposal volume is acceptable; however, the Offeror shall ensure that all acronyms are defined in the glossary for that volume. These pages will not be included in the page count for any volume.

4.1 Volume I – Governance

Volume I shall not exceed the pages limits identified in the table above. Any pages in excess of the allowable content, other than those specifically excluded from the overall page limit, will be removed from the Offeror's proposal and not evaluated.

4.1.1 Executive Summary

The Offeror shall present their proposed system governance solution for the T3TLv program, including their team construct and expected contributions by each Major Teammate. The Executive Summary should include a summary of the Offeror's programmatic and systems engineering approaches, to include the cost and risk management approaches.

4.1.2 Program Management

The Offeror shall detail the people, processes, and tools planned to manage and conduct the program to a successful completion on time and within the agreed upon price. The Offeror shall discuss their program, personnel, cost, and risk management approaches, including any processes and tools. The Offeror shall include a list of the primary technical and schedule risks on the project, the planned mitigations, and the approach for managing risks throughout the system lifecycle. Risks and mitigations should be focused on execution of the proposed effort. Decisions made while developing the proposed solution (e.g., investments made prior to proposal submission, etc.) are not appropriate to cite as risks or mitigations. Schedule management shall be discussed in Volume III; however, the Offeror is free to cross-reference Volumes I and III if doing so will be of benefit in explaining the value and quality of their approach.

The Offeror should describe processes to staff the program with qualified personnel, identify and manage critical vendors and subcontractors, and control costs within a Firm Fixed Price (FFP) environment. The Offeror shall describe the schedule and performance risk associated with other commitments by the Prime and subcontractor teams and their facilities during the period of performance, including, but not limited to, commitments to other SDA programs. If the Offeror supports other SDA programs, the Offeror shall clearly define how they will support T3TLv with minimal disruptions to their existing SDA programs, inclusive of personnel and facilities.

The Offeror shall describe their approach to retaining or obtaining appropriately cleared staff and facilities to support classified work at levels up to TOP SECRET/SCI.

The Offeror shall document planned communications with the Government, detailing the purpose, entrance criteria, key accomplishments anticipated, exit criteria, and documentation methodologies for each meeting, review, or key management milestone within the program.

The Offeror shall plan the program using the program reviews identified in Section 2.1 of Attachment 1 – Statement of Work and in accordance with other dates specified throughout Attachment 1.

The Offeror shall explicitly state the need for any anticipated waivers to Government standards and/or compliance documents identified in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document.

4.1.2.1 Subcontractor and Supply Chain Management

The Offeror shall address their subcontractor and supply chain management methodologies. The Offeror shall include a table showing the delivery schedule of critical parts. The Offeror shall present a strategy for managing delivery performance of major subcontractors with finite production capacity.

The Offeror should describe how they will minimize the effect on the program from fluctuations in supply chain availability and subcontractor deliveries. The Offeror should propose a contingency plan in response to supply shortages for all critical parts to include pricing, supply, and delivery information.

The Offeror shall include a sparing plan for critical and long lead parts/units addressing acquisition of critical spares, inventory sources, and associated risks.

The Offeror should provide an assessment of hardware, firmware, and software maintainability, including the availability of manufacturer/developer support for all elements of the design to ensure that enduring product support for security patches and maintenance support remains available over the operational life of the system.

4.1.2.2 Nontraditional Defense Contractors (NDC) / Nonprofit Research Institution

If applicable, the Offeror shall identify which of their subcontractors constitutes an NDC or a nonprofit research institution in accordance with 10 U.S.C. 2302(9). The Offeror shall identify the extent of participation of NDC(s) and/or nonprofit research institution(s) in the Offeror's T3TL program.

4.1.3 Security Performance

The Offeror shall describe the Offeror's security approach and ability to comply with the security requirements of this solicitation. It should contain the Offeror's security approach, the approach to meeting the clearance requirements for proposed personnel, facility locations, and classified information safeguarding.

4.1.3.1 Facilities Clearance

The Offeror shall identify its proposed location(s) for performing classified efforts for this program. The Offeror is required to hold a current Top Secret facility clearance issued by DCSA in accordance with 32 CFR Part 117 and must provide the complete address and CAGE Code of the facility using

Attachment 7a – Personnel & Facilities Template.**4.1.3.2 Key Personnel Clearances**

The Key Personnel listed in section 3.1.1.8 MUST have current, Top Secret (TS) clearances with SCI eligibility and be available to work at contract start. The Offeror shall provide the required information for the Key Personnel listed in section 3.1.1.8 using Attachment XX – Personnel & Facilities Template.

4.1.3.3 Foreign Ownership and Controlling Interest

The Offeror shall describe its approach for satisfying all requirements of this solicitation related to Foreign Ownership, Control, or Influence (FOCI). The Offeror's security approach should identify, at a minimum, the following:

Compliance with Disclosure of Ownership or Control by a Foreign Government (DFAR 252.209-7002) and shall include submission of a current SF 328, "Certificate Pertaining to Foreign Interests (FOCI)" if applicable. Offerors must provide the same information for each proposed subcontractor or subsidiary if applicable. If Offeror's are foreign owned, they must have an approved FOCI mitigation plan and/or a Cognizant Security Agency (CSA) Security Control Agreement (SCA), or Special Security Agreement (SSA). For Companies that will require access to Collateral, SCI, or COMSEC information, a National Interest Determination (NID) is required.

4.1.3.4 Contract Security Classification Specification (DD254)

The Offeror shall complete DD254 and submit it with the Security Volume. The DD254 should include the CAGE code and address of all locations where classified work is anticipated.

4.1.4 Facilities Management

The Offeror shall detail the secure facilities that will be used to execute the T3TLv program. The Offeror may reasonably assume that SDA will support co-use arrangements but should not assume that facilities can be newly certified on a schedule suitable for supporting this program. The Offeror shall include a table listing the primary facilities or rooms to be used. For each item, the Offeror shall provide the area, the maximum classification level, the certification status, the certifying organization, and networks and voice communications capabilities available for use. The Government expects that regular access to classified networks is important for execution of this program.

4.1.5 Systems Engineering

The Offeror shall detail the systems engineering processes and methods that will be used to execute the T3TLv program. The Offeror shall clearly identify their approach to systems engineering and detail the use of any processes and tools that will be used on the program. The Offeror shall identify any reference material, such as corporate command media or published standards, that are used as the basis for their systems engineering approach and discuss how such material will be tailored for the T3TLv program. The Government expects that the Offeror will utilize systems engineering best practices.

4.1.6 Mission Assurance

The Offeror shall describe the mission assurance approach and Offeror's quality systems. The Offeror should include a discussion of relevant standards and certifications. In particular, the

Government is interested in how the Offeror plans to tailor – if applicable – existing mission assurance processes and tools to meet the requirements of a proliferated LEO constellation designed for a 60-month operational mission life lifetime.

4.1.7 Organizational Conflict of Interest

The Government has determined there is potential organizational conflict of interest due to the nature of the anticipated separate solicitations and contracts for the entire Tranche 3 Program. The entire, fully integrated T3 program will likely include multiple T3 SV contractor(s)/performers (primes and subs) and T3 Ground contractor(s)/performers (primes and subs) who will interface with the T3TLv Performer. Therefore, if a T3TLv Performer intends to bid or perform on any other T3 contract/agreement, there may be an OCI. In addition, the Space Development Agency's OCI policy clearly states that SDA SETA must remain OCI free within SDA; therefore, a contractor cannot concurrently be an SDA SETA contractor and a T3TLv Performer.

As a part of the proposal, the Offeror shall provide the Agreements Officer with complete information of previous, potentially impending, or ongoing work that is in any way associated with this acquisition. The Offeror shall provide a written assessment addressing why there is no OCI and whether a future OCI is likely or will be avoided. In the event there is a known OCI or if the bidder intends to bid on future T3 Program acquisitions, the Offeror shall include an OCI Plan in addition to the Assessment. An Offeror who assesses they will avoid OCI with the T3 Program MUST include a statement (guarantee) attesting they will avoid OCI for the life of this T3TLv Agreement. If no OCI exists or will exist, the Offeror need only provide the Assessment. If a conflict exists or may exist, both the Assessment and the OCI are required to be a complete proposal.

The resulting Agreement will include an organizational conflict of interest limitation applicable to the related Government scope of work, at either a prime contract level, at any subcontract tier, or both. During evaluation of proposals, the Government may also, after engagement with the Offeror and consideration of ways to avoid the conflict of interest, insert a special provision addressing eligibility for future contract awards within the existing Tranche 3 Program. The OCI mitigation plan will not be included in the Volume I page limit.

4.1.8 Key Personnel Résumés

Key Personnel résumés shall not exceed two (2) pages each using Attachment XX – Resume Template. The personnel résumés shall be included as attachments to this volume and will not be included in the Volume I page limit. Classified personnel résumés may also be submitted as attachments and will not be included in the Volume I page limit.

The Offeror shall identify the Key Personnel planned to be assigned to this program from the Offeror and Major Teammates as appropriate. Résumés should be in bullet format and include:

- Summary statement highlighting experience and expertise
- Current role with the Offeror or teammate to include current assignments and programs with particular attention paid to efforts currently underway for SDA to include percentage of time already applied to other SDA programs or projects
- Planned role within the T3TLv program
- Evidence of applicable mission experience or understanding
- Education (academic and military, if applicable)
- Anticipated percentage of time to be assigned to the T3TLv program
- Signed statement agreeing to support the SDA T3TLv program if awarded

Proposed key personnel not currently employed by the Offeror may be included as appropriate. In those cases, the Offeror shall include signed contingent offer letters. These letters should be signed by both the Offeror and the proposed staff member and should state the willingness of the proposed staff member to join the Offeror in support of the SDA T3TLv program if awarded.

At a minimum, key personnel shall include the staff supporting the proposed program in the following positions:

From the Offeror:

- Program Manager
- Chief Engineer
- IBS-L/TACSATCOM Mission Payload Lead(s)
- Assembly, Integration, and Test (AI&T) Lead
- Subsystem Lead(s) (if Offeror is developing any of the major subsystems. The major subsystems are those listed in the Major Teammate definition in Section 3.)
- Information Security Certified Professional

From each Major Teammate:

- Program Manager
- Chief Engineer

4.1.9 Personnel Matrix

The personnel matrix shall be submitted using Attachment 7c – Personnel Matrix as an attachment to this volume and will not be included in the Volume I page limit.

4.1.10 Technical Data and Computer Software Rights Assertions

The technical data and computer software rights assertions shall be submitted as an attachment to this volume and will not be included in the Volume I page limit.

The Offeror shall clearly delineate **all** limitations on data, hardware, and computer software rights for technical data, hardware, or computer software provided by the Offeror and **all of its teammates**, in the format shown in Attachment 4 of the Draft Other Transaction Agreement (OTA). It is the Government's assumption that **no** hardware, technical data, or computer software will be provided with less than Government Purpose Rights and that the vast majority of information generated will be provided with Unlimited Rights to the Government. The Offeror shall clearly identify those elements to be provided with less than Unlimited Rights and identify the rationale for their assertion(s) using the format shown in Attachment 4. Any item related to the performance of this effort and not contained in Attachment 4, shall be delivered upon request to the Government, with no less than Government Purpose Rights. The terms "Unlimited Rights", "Government Purpose Rights", "Limited Rights", and "Restricted Rights" are defined in Volume VI Draft Other Transaction (OT) Agreement.

4.2 Volume II – Technical

Volume II shall not exceed the pages limits identified in the table above. Any pages in excess of the allowable **content, other than those specifically excluded from the overall page limit, will be removed from the Offeror's proposal and not evaluated.** Up to ten (10) of the thirty-five (35) pages in the primary Volume II submission may be classified, and the remainder of each primary

Volume II submission must be unclassified. With the exception of Commercial Data Sheets, Volume II appendices may also be classified.

4.2.1 Executive Summary

The Offeror shall present their proposed solution concept for two (2) planes of T3TL_v SVs and its NOVA ground system. The Executive Summary shall include an Operational Viewpoint (OV)-1 or equivalent high-level overview of the proposed solution. The graphics shall depict the SVs, their payloads, the means by which they communicate with each other and the ground, and the means by which they provide data to the Warfighter. The Offeror's milestone-driven integrated master schedule for capability delivery, culminating in an in-orbit delivery and performance per SDA's schedule requirements, shall be integrated into this graphical representation.

4.2.2 CONOPS Overview

The Offeror shall provide a CONOPS for the operation of the T3TL_v constellation, including network maintenance and initialization, data transfer, and direct-to-Warfighter communications. The Offeror shall clearly identify any mission constraints or limitations.

4.2.3 Space Vehicle (SV) Overview

The Offeror shall present an overview of the SV, including but not limited to:

- SV block diagram
- Delta-V budget
- High-level mass budget
- Power budget with power modes
- Vehicle configuration showing all major components
- Backup TT&C system design

This section should also include an overview of the SV concept of operations (CONOPS).

A detailed Master Equipment List (MEL) with mass, power, Manufacturing Readiness Level (MRL), and Technology Readiness Level (TRL) or other flight heritage shall be included as an appendix and will not count against the Volume II page limit. The MEL is expected to include details of assemblies below the subsystem level, including a Bill of Materials and Materials in use List where available. The power budget shall include the duty cycle and current best estimate for each component by power mode, as well as the orbit average power generated in each power mode and enough information to determine the battery depth of discharge.

4.2.3.1 T3TL_v Payload Architecture and Design

The Offeror shall present their overall payload architecture and design. For each payload element below, the Offeror shall include, but not be limited to, a discussion of payload hardware, software, control, data transfer capability and/or limitations, and power requirements. Proposals shall demonstrate a clear understanding of each payload's mission application and the technical solution's ability to accomplish that mission. Proposals shall also address any needed test and certification requirements and plans and shall include schedule and risk assessments for subject completion. The Offeror shall detail their trade-space analyses for critical payload subsystems, including power amplifiers, software-defined radios, and antennas.

4.2.3.2 IBS-L Mission Payload

The Offeror shall provide a detailed description of their IBS-L mission payload. The Offeror shall

provide detailed payload performance parameters, demonstrating how the IBS-L mission payload meets system requirements and satisfies its mission application.

The Offeror shall describe how the mission payloads are integrated with the bus and how IBS-L traffic will be managed. The Offeror shall describe how IBS-L mission payload functions and parameters are configured and commanded by the Performer's NOVA, including transmit power, center frequency, and data rate settings. The Offeror shall describe their approach to commanding and managing 1- and 2-channel modes of IBS-L mission payload operations.

The Offeror shall address the proposed design of the entire baseline IBS-L mission payload system, including antennas, RF hardware, and transmitters, as well as power requirements, thermal requirements, reliability in the radiation environment, shock/vibe compliance, and Tx/Rx isolation.

The Offeror shall describe their approach to meeting signal quality specifications in each mode of IBS-L mission payload operations, including a description of planned test and intermodulation analysis to ensure that no combination of center frequencies for the 2-channel IBS-L mode violates specifications. This approach shall include a discussion of analyses that will be performed to predict the frequency and amplitude of linear and non-linear distortion products, such as harmonics, spurs, and intermodulation products.

The Offer shall describe how broadcast message and fill data will be managed to ensure the broadcast stream is free of discontinuities and yet minimizes latency of message broadcast and prevents latency buildup.

4.2.3.3 TACSATCOM Mission Payloads

The Offeror shall provide a detailed description of their S-band TACSATCOM and UHF TACSATCOM mission payloads. The Offeror shall provide detailed payload performance parameters, demonstrating how each payload meets system requirements and satisfies its mission application.

The Offeror shall describe how the mission payloads are integrated with the bus and how TACSATCOM traffic will be managed. The Offeror shall describe how TACSATCOM mission payload functions and parameters are configured and commanded by the NOVA, including transmit power, center frequency, and PN code settings.

The Offeror shall address the proposed design of the entire baseline TACSATCOM mission payload system, including antennas, RF hardware, demodulators, forward error correction (FEC) and cyclic redundancy check (CRC) implementations, as well as power requirements, thermal requirements, reliability in the radiation environment, shock/vibe compliance, and Tx/Rx isolation.

4.2.3.4 Optical Communication Terminals (OCT)

The Offeror shall provide a detailed description of the OCTs, including their placement on the SV, individual and total fields of regard, and ability to create and maintain space-to-terrestrial and space-to-space links. The Offeror shall identify the anticipated power requirements, data rate, bit error rate, and any other parameters considered key by the Offeror. The Offeror should provide a link budget demonstrating the closure of the optical links and clearly state any assumptions used in the development of the link budget. The Government expects that the link budgets will be developed using industry standards for margin (e.g., 3 dB). The Offeror shall identify the compliance of their current system with the SDA OCT Standard v3.2, detail any non-compliant elements of their design, and describe the path to full compliance for those elements.

4.2.3.5 Ka-band Mission Payload System

The Offeror shall provide a detailed description of the K_a-band communications mission payload. The Offeror shall identify the anticipated modulation and coding implementations, information rates, power requirements, bit error rate, and any other parameters key to mission performance. The Offeror shall identify the antenna configurations, field of regard, field of view, and deployment mechanisms for each of the antenna systems. The Offeror shall describe how the payload implementation complies with the requirements to concurrently support multiple tactical users. The Offeror shall provide link budgets demonstrating the closure of required RF links and clearly state all assumptions used in the development of the link budgets. The Government expects that the link budgets will be developed using industry standards for margin (e.g., 3 dB). The Offeror shall include a link closure analysis that substantiates the payload's compliance with the requirement to concurrently support multiple tactical users over the required footprint under the specified link conditions.

4.2.3.6 Telemetry, Tracking, and Control (TT&C)

The Offeror shall provide a detailed description of the RF communications subsystems used for telemetry, tracking, and control (TT&C). The Offeror shall identify the anticipated modulation and coding implementations, information rates, power requirements, bit error rate, and any other parameters key to mission performance. The Offeror shall identify the antenna configurations, field of regard, and field of view supporting link closure at the required information rate, including the impact of blockages. The Offeror shall provide link budgets demonstrating the closure of required RF links and clearly state all assumptions used in the development of the link budget. The Government expects that the link budgets will be developed using industry standards for margin (e.g., 3 dB).

4.2.3.7 Networking and Data Routing

The Offeror shall describe the networking capabilities enabled by their technical solution. The Offeror should detail the compliance of their system with the SDA NEBULA Standard v3.06, identify any non-compliant elements of their design, and identify the path to full compliance for those elements. The Offeror should discuss their plans for on-orbit network initialization and provide estimates of system performance.

4.2.3.8 Red-Side Processing (RSP)

The Offeror shall describe the proposed approach to implementing a red-side processing capability to host dynamic network management applications, including which processing unit the capability is expected to be hosted on, the processing, memory, and storage resources reserved for hosted applications, and a description of how this capability meets all functional and interface requirements.

4.2.3.9 GNSS SA Mission Payload

The Offeror shall describe their proposed GNSS SA mission payload implementation and its expected performance. The Offeror shall describe its planned implementation of a circular storage buffer and the process of collecting, storing, retrieving, and downlinking data products as commanded.

4.2.4 Space Vehicle (SV) Bus Design

The Offeror shall present their SV design based on the manufacture or purchase of commoditized SV buses¹. Should the Offeror elect to propose a non-commoditized bus alternative, they shall

¹ A commodity space vehicle bus utilizes an existing design that can be easily manufactured and reused across multiple

present their process for commoditizing the alternative bus in addition to detailed justification for their bus selection.

4.2.4.1 Bus Design and Heritage

The Offeror shall provide a detailed description of the SV bus including the size, weight, and power available. The Offeror shall present a discussion of the bus heritage to include:

- Where and how many have flown
- Design life and experienced flight life
- Types and frequency of experienced anomalies and their solution(s)

If the proposed bus has not flown, the Offeror shall provide the heritage from an existing bus (if applicable) and the changes made to reach the proposed bus design along with the rationale for each proposed change. Risk reduction activities and planned qualification efforts will be provided.

4.2.4.2 Space Vehicle (SV) Subsystems

For each of the major subsystems (e.g., electrical and power, attitude control, TT&C, etc.), the Offeror shall provide a detailed description of the subsystem, as well as system heritage (similar to that for *Bus Design and Heritage* above). Block diagrams are encouraged where they provide greater insight into the design approach. The Offeror shall discuss specialty engineering (e.g., radiation, electromagnetic interference, contamination control, etc.) issues and approaches as appropriate. The Offeror should discuss the overall fault management approach, including the safe mode implementation.

4.2.4.3 Cybersecurity

The Offeror shall discuss their approach to implementing required Risk Management Framework (RMF) controls and cybersecurity in their integrated space and ground system.

4.2.4.4 Encryption

The Offeror shall describe the proposed implementation of security concerns, such as red-black isolation, encryption, and command authentication. The Offeror shall describe the encryption systems and technologies used on the space vehicle and how these systems will obtain certification and interoperate with the systems of other PWSA vendors and the ground.

4.2.4.5 Navigation

The Offeror shall describe the proposed implementation of a resilient navigation subsystem that is compliant with all functional and performance requirements.

4.2.5 Manufacturability

The Offeror shall describe their manufacturing and assembly processes, their AI&T flow and how it supports the on- time delivery of the T3TL^o SVs. The Offeror should explain their ability to procure or manufacture commoditized buses at low cost and high reliability while achieving economies of scale as demonstrated in the commercial sector. The Offeror shall further detail planned mitigations for any manufacturing-related risks cited in Volume I, including strategies for overcoming the inability of the Offeror or a Major Teammate to produce the required number of units. The Offeror

SVs and for multiple missions. A commoditized bus has been designed for manufacture and is suitable for, or currently being produced by, an assembly line-like process at scale.

shall provide estimates of the manufacturing capacity of their organization (if applicable) and any Major Teammate, as identified in Section 3, inclusive of current contracts and any potential contracts within the Tranche 3 timeframe.

If currently under agreement or performing scope for SDA, the Offeror shall clearly and concisely document the ability to produce T3TLv SVs, to include all payloads, within manufacturing constraints imposed by ongoing SDA efforts.

4.2.5.1 Supply Chain

The Offeror shall address production capacity, supply chain capacity, supply chain security, and production scalability issues for the T3TLv SV buses and key payloads including, but not limited to, OCTs, IBS-L, TACSATCOM, and cryptographic equipment.

4.2.6 Test, Demonstration, and Verification

The Offeror shall present their hardware and software, ground and orbit-based test and demonstration approach(es). This section should address all aspects of test and demonstration to include modeling, simulation, analysis, emulation, and ground- and space-based testing. The Offeror should address requirements verification at the component, subsystem, system, vehicle, and constellation levels up to and through environmental tests. The Offeror shall describe test approaches, methodologies, performance, verification, documentation, and staffing. This section should include sample test artifacts from previous programs, and identification of test platforms and hardware maintained by the Offeror. For cases where test activities are expected to be accomplished outside of the Offeror's direct control, the Offeror should identify their approach to manage testing and identify the experience level of the organization the Offeror plans to use to conduct the test. Of particular interest to SDA is the Offeror's approach to conducting tests and demonstrations prior to launch and during early orbit operations and checkout, as well as use of the SVs in conjunction with Warfighter exercises once in orbit. Throughout discussion of their test activities, the Offeror should clearly identify live, virtual, and constructive tests and demonstrations, if applicable, and any elements expected to be provided by the Government to facilitate completion of their verification program.

4.2.7 Launch Services Overview

SDA will procure launch services using the NSSL program. The Offeror shall assume each plane of ten (10) T3TLv SVs will be launched on a single, Evolved Expendable Launch Vehicle-Medium (EELV-M) class launch vehicle. The Offeror shall describe how twenty (20) of their SVs will integrate onto a single EELV-M class launch vehicle, including any custom adapters or separation required to attach to the launch vehicle. The Offeror should include graphics of all twenty (20) vehicles integrated onto a representative launch vehicle. The Offeror shall clearly identify any limitations that their design imparts on the launch vehicle and shall identify the total mass of the SVs and any adapter systems and/or leave behind mass. The Offeror shall identify any launch services (such as a T-0 dry nitrogen purge) or additional accesses required to support launch vehicle integration. If the proposed bus has flown on an NSSL class launch vehicle, the Offeror shall provide the bus heritage. For pricing purposes, the Offeror shall assume that all launches will be conducted from Cape Canaveral Space Force Station, FL.

4.2.8 Ground Segment Integration and Operations

The Offeror shall present their approaches to operating their constellation through their NOVA software. The Offeror shall present their approach to integrating their NOVA with the Government-

provided SUPERNOVA. The Offeror shall define which functions their NOVA will provide, which functions are assumed to be provided by SUPERNOVA, and how the systems will interact during operations and maintenance activities. The Offeror shall describe their use of COTS software and tools and the resultant benefits to integrated operations. The Offeror shall describe how their ground system will command SVs and provide SV system health and status data to the integrated ground operations management systems for continuous situational awareness and performance monitoring.

The Offeror shall describe the proposed design of their software-based NOVA system to include descriptions and functions of all external interfaces, including those that enable connectivity to the Offeror's factory. The Offeror shall identify the minimum requirements for the GFE hardware or cloud-based service needed to host their proposed NOVA system to include the number of cores, amount of RAM, and amount of storage that must be available from a Government-provided hosting service or platform.

The Offeror shall describe their approach to developing and maintaining the SV Operational Capabilities and Limitations (SOCAL) model to ensure that SUPERNOVA has ready access to an accurate and current representation of the Offeror's SVs' performance and operational constraints. The Offeror shall describe how the SV NOVA will comply with its requirements for responsive mission plan feasibility assessments and responsive translation of approved mission plans to SV commands and subsequent upload of the commands for execution.

The Offeror shall describe T3TLv NOVA interaction with SUPERNOVA to route IBS-L and TACSATCOM messages earmarked for different geographic regions around the world to the correct SV at the correct epoch to ensure uninterrupted service to terrestrial receivers. Descriptions shall include handoff management from rising and setting SVs such that no messages are dropped and the use of broadcast frequencies that preclude interference where geographic regions overlap. The Offeror shall explicitly identify which subsets of this capability will be provided by NOVA and which are expected to be provided by SUPERNOVA.

The Offeror shall describe the suggested approach for verification that the proposed SVs and NOVA are compatible with SUPERNOVA and the greater PWSA GS.

The Offeror shall describe the proposed design of their FlatSat(s) and their planned use in support of system verification, test, and anomaly resolution.

4.2.9 Mission Operations, Sustainment, and Disposal

The Offeror shall identify the support that they will provide to T3TLv operations during nominal constellation operations, surge support, and anomaly operations. The Offeror shall describe their operations concept, level of automation, and staffing requirements for nominal operations. The Offeror shall describe their approach for anomaly resolution, deficiency reporting, and planned and unplanned system maintenance. The Offeror should list the documentation and training anticipated to be provided to SDA operations personnel. The Offeror shall describe the disposal plan for each SV and how that plan will reliably dispose of decommissioned SVs.

4.2.10 Appendices

The following appendices shall be included as attachments to Volume II, subject to the page limitations as indicated, and will not be included in the Volume II page limit.

4.2.10.1 TACSATCOM Payload Design Technical Volume Response

Details of the TACSATCOM Payload Design Technical Volume Response are provided in the

T3TLv Program Solicitation Appendix.

4.2.10.2 Multi-Carrier IBS-L Mission Payload Design

Describe IBS-L system engineering and payload design solutions that simultaneously provide two separate carriers at specified EIRP across the elevation angles per TRD, while maintaining compliance to the adjacent channel emission (ACE) requirement in MIL-STD-188-181C Table II. All shall be within reasonable Size weight and power for the proposed bus.

Describe expected amplifier and ACE performance to include but not limited to amplifier saturation power, P_{in} vs. P_{out} , AM/AM, AM/PM curves, intermodulation levels vs backoff power and number of channel and channel spacing combination over the transmit frequency requirement band.

Also describe expected Peak to average power ratios in the amplifier(s) and any resulting restrictions on carrier spacing or carrier phase relationships or carrier data/symbol limitations.

Show the final amplifier design trades. Options may include but is not limited to:

- Linearization or Predistortion Techniques
- Differing Amplifier Topologies
- Band limitations
- Band Channelizing with band pass filters
- Limitation of number of simultaneous carriers
- EIRP relief
- Using multiple amplifiers and combiner

4.2.10.3 State of Charge Analysis

Details of the State of Charge Analysis are provided in the T3TLv Program Solicitation Appendix.

4.2.10.4 Mission Availability and Reliability

The T3TLv Technical Requirements Document includes notional requirements for the availability and Mean Time to Recovery (MTTR) of the networking mission functionality. The Offeror shall provide a detailed, quantitative assessment of the performance of their proposed design and the features that enable that performance. If the Offeror wishes to propose other availability and reliability requirements, those rationale for those requirements shall be supported with analysis demonstrating the mission impacts that result from the proposed modifications. The Offeror shall also propose an approach for verifying the availability and reliability requirements prior to launch, as well as maintaining and updating availability performance estimates on orbit.

4.2.10.5 IBS-L and TACSATCOM Mission Payload RF Link Budgets

The Offeror shall submit comprehensive RF link budgets covering the full fields of regard for each of the IBS-L and TACSATCOM mission payloads. The Government has provided required effective isotropic radiated power (EIRP) values for transmit functions and required performance specifications at specified received signal strength indicator (RSSI) values for receive functions. The Offeror's link budgets need not consider parameters specific to tactical terminal performance because those parameters have been factored into the specified EIRP and RSSI requirements. The Offeror's link budgets shall include estimated losses due to Doppler for receive functions.

This appendix is expected to include link budget tables alone and has no page limit. Any normal text submitted within this appendix will be discarded. Any attempts to circumvent these instructions will

be removed from the proposal and not evaluated.

4.2.10.6 Launch, Early Operations, Commissioning, and Transition to Operations Plan

The Offeror shall describe in detail their approach to LEOPs, verification and functional checkout, and transition of operations to the OCs. Specifically, the Offeror shall include:

- Detailed schedule of activities
- Entry and exit criteria to successfully demonstrate commissioning objectives
- Roles and responsibilities of the Offeror, GS integrator, and the Government
- All resources used in support of these activities
- All Government-furnished equipment/information/facilities/personnel (GFE/I/F/P) and assumptions
- Time-phased staffing plan
- Relevant experience with the proposed approach
- Significant technical risks and mitigation strategies for the proposed approach

This appendix shall not exceed five (5) pages.

4.2.10.7 Commercial Data Sheets

The Offeror shall submit commercial data sheets for any major subsystem identified as COTS, as defined in Section 3. Additionally, the Offeror may submit up to ten (10) commercial data sheets for other systems, subsystems, or components.

Commercial data sheets shall be submitted as a list of hyperlinks to publicly available, downloadable .pdf data sheets on the Offeror's websites or their subcontractors' websites. Each downloadable .pdf file shall be no more than two (2) pages. Links will be accessed in the order provided in the proposal and only the first twenty (20) pages beyond those associated with major subsystems cited as COTS will be considered. Any pages beyond the twenty (20)-page limit will be discarded.

4.2.10.8 Parameter Specification

The Offeror shall complete the value column in each worksheet of the T3TL-UpsilonKeyParameterSpecification.xlsx file in the Bidder's Library and shall include the completed .xlsx file as an appendix to Volume II. This file shall be submitted as a Microsoft Excel (.xlsx) file. Parameter specifications submitted as .pdf files will be discarded. All values in the table shall be provided in the units specified. Upon selection for award and at the discretion of the Government, the proposed values in this table may be included in the final Statement of Work and Technical Requirements Document in lieu of the current threshold and objective values specified in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document.

4.2.10.9 Technical Requirements Verification Methods

The Offeror shall submit a version of Attachment 3 – Technical Requirements Document as an appendix to Volume II, with a column appended to specify the Offeror's proposed verification method for each requirement. If the proposed verification method differs from the Government's recommended verification method, the Offeror shall include rationale for the proposed alternative method in an adjacent column.

4.3 Volume III – Integrated Master Schedule

Volume III shall not exceed the pages limits identified in the table above. Any pages in excess of the allowable **content, other than those specifically excluded from the overall page limit, will be removed from the Offeror’s proposal and not evaluated.**

4.3.1 Integrated Master Schedule (IMS)

The Offeror shall present the milestone-driven IMS designed to meet the Government’s notional milestones provided in Section 2.1 of Attachment 1 – Statement of Work and launch schedule identified in Section 4 of Attachment 1 – Statement of Work. The Offeror shall clearly indicate the anticipated schedule margin for each launch, where margin is calculated as the number of calendar days between the date of completion of the final SV for the launch and the corresponding delivery date to the launch site, as specified in Attachment 1 – Statement of Work. The Offeror shall identify the dates for all major reviews and program milestones and include entrance criteria, key accomplishments, and exit criteria for each milestone. If the milestones proposed correspond to payments anticipated by the Offeror, this Volume should so state and link to the appropriate section in *Volume IV – Price and Rationale* so that the milestone payment appears in *Volume III – Integrated Master Schedule* and the payment amount, terms, conditions, and Government acceptance criteria for payment appear in *Volume IV – Price and Rationale*.

At least two (2) of the pages shall be a separable, standalone detailed Gantt chart (or similar illustration) clearly presenting the Offeror’s milestone-driven integrated master schedule, including key program milestones and critical paths. This chart should include dependencies, assumed Government or Major Teammate hardware, software, or information deliveries and/or approvals, tests, demonstrations, reviews, and/or any other key activities in the Offeror’s proposed program.

The Offeror shall clearly delineate the primary and secondary critical paths within their schedule and any additional critical paths deemed appropriate to justify their ability to deliver the proposed SVs on time and on cost. The Offeror shall discuss the critical path analysis used to determine the primary and secondary critical paths. The Offeror shall discuss their approach to managing program progress against the baseline schedule. The Offeror shall include delivery dates of components from all Major Teammates and shall clearly identify which of these components lie on the critical path.

The Offeror shall identify maturation and risk reduction plans for any SV components that are at a TRL of less than six (6) at the time of the proposal. Key technical events in the maturation of these technologies should be clearly presented in the IMS appendix. The Offeror shall elaborate upon their approach(es) to mitigate schedule risks identified in Volume I and ensure on-time delivery. The Offeror shall identify which teammate deliveries and which external dependencies produce the most risk to on-time delivery.

The Offeror shall deliver their complete IMS in Microsoft Project-compliant native format, as well as in .pdf format in conjunction with this volume. The complete IMS shall include schedule details down to at least Level 3 of the work breakdown structure (WBS) and details to at least the subassembly level for high-risk areas and to the subsystem level elsewhere. IMS content shall be tailored to address each WBS element’s unique design, development, assembly, integration, and test activities, as applicable.

The delivered IMS files (Microsoft Project and .pdf format) will not count against the Volume III page limit, however, the Gantt chart is included in the page limit.

4.3.2 Schedule Risk

The Offeror shall clearly and concisely identify schedule risks based on their approach, design, team construct, and/or consolidated backlog. For each risk, the Offeror shall identify key mitigation steps as well as internal and external dependencies of those steps.

4.3.3 Appendix – Portfolio IMS

If the Offeror is currently providing SVs, ground systems or other materiel to SDA, they shall submit a Portfolio IMS as an Appendix to Volume III. This Portfolio IMS shall not exceed five (5) total pages and will not count against the Volume III page limit.

At least two (2) pages shall be a separable, standalone detailed Gantt chart (or similar illustration) clearly presenting the Offeror's milestone-driven portfolio-level integrated master schedule, including key program milestones and critical paths. This chart should include dependencies, assumed Government or Major Teammate hardware, software, or information deliveries and/or approvals, tests, demonstrations, reviews, and/or any other key activities in the Offeror's proposed SDA portfolio. The Offeror shall identify portfolio-level schedule risks, their mitigations, and succinctly discuss dependencies of one program upon another and/or how each program will be completed successfully even if another program in the portfolio encounters schedule, technical or programmatic difficulties.

4.4 Volume IV – Price and Rationale

4.4.1 Executive Summary and Pricing Rationale

The Offeror shall detail their pricing assumptions and rationale that guided their Firm Fixed Price (FFP) submission. The Government needs to understand the Offeror's assumptions in building their final price such that any discussions deemed of value can be conducted. The Offeror should discuss any constraints on the Government such as time to review and/or approve contractual documentation, constraints on mission partners such as time to receive needed data or information, etc. Where appropriate, the Offeror should discuss their use of learning in driving down subsequent SV costs when producing more than a single copy and where such learning has led to cost savings on previous programs.

4.4.2 Price Proposal Structure

The Offeror's price proposal shall comply with the guidance in the following sections. The Offeror may assume an anticipated Authorization to Proceed (ATP) of 31 July 2025.

4.4.2.1 Bidding Unit

The Offeror shall provide the price for one (1) full unit, where a unit consists of:

- Two (2) orbital planes of ten (10) T3TL_v SVs (20 total T3TL_v SVs) and system-level program management and systems engineering T3TL-Upsilon
- SV to LV integration
- All Approved Development Units (ADU) for interoperability testing
- NOVA and integration into GS
- FlatSat
- Two (2) years of operations and sustainment following the second plane's Functional Acceptance Review (FAR) as part of base contract

- Three (3) one (1) year options of operations and sustainment following the base period of this agreement
- All other efforts necessary to meet the requirements as specified in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document

4.4.2.2 CLIN Structure

The Government anticipates awarding to a single CLIN. The CLIN will be awarded as Firm Fixed Price (FFP). All price details of the CLIN shall be included for a compliant proposal. The Offeror shall propose milestone payment structures in accordance with the instructions in the following sections. This information will be used by the Government to evaluate the reasonableness of, and risks associated with, the proposed technical solution, schedule, and price.

4.4.2.3 Work Breakdown Structure (WBS)

The Offeror shall provide a WBS, compliant to the Level 3 WBS shown in Figure 3-1. The Offeror may include, at their discretion, work breakdown details below the levels indicated.

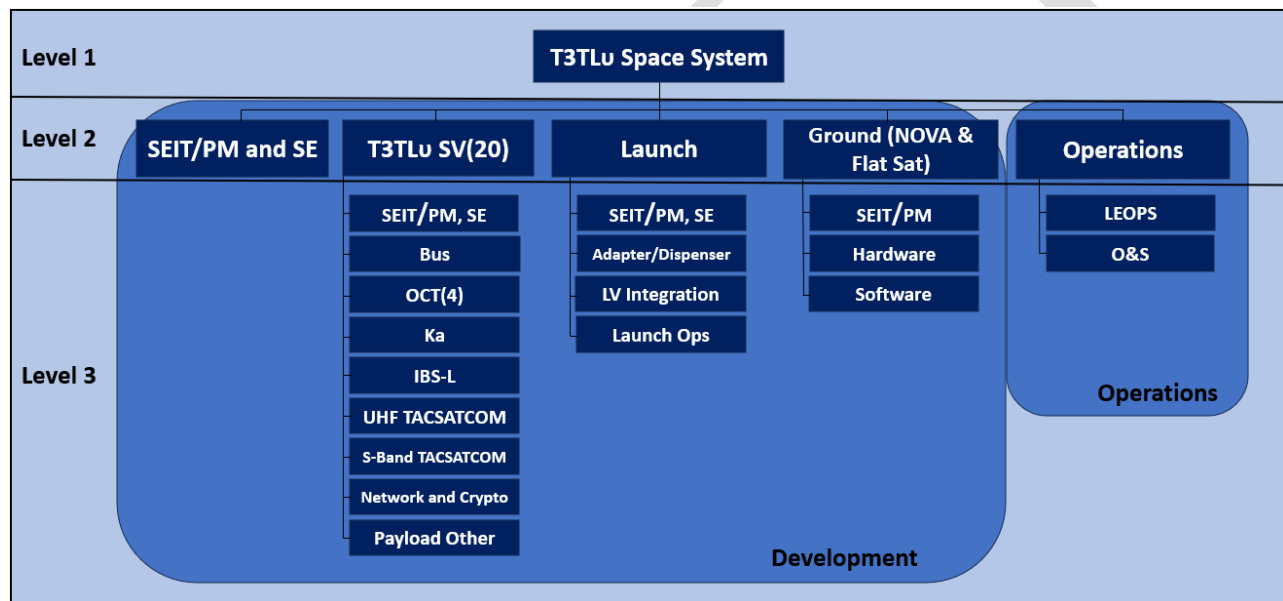


Figure 3-1 – Level 3 WBS

4.4.2.4 Milestone Payments

The Government intends to fund this FFP effort via event-based Milestone Payments. The Offeror shall propose a Milestone Payment Plan compliant to guidance in this section.

The Offeror shall propose a milestone payment structure using Attachment 2 Schedule of Milestone Payments (MPP). The Offeror's payment plan shall adhere to mandatory milestones and minimum percentages listed in 3-3 but may otherwise add, delete, or modify in areas where not specified.

Table 3-3 – Milestone Payments

Milestone	Mandatory	Minimum % of Total Price
Kickoff	No	—
System Requirements / Design Review (SRR/SDR)	Yes	5%

Preliminary Design Review (PDR)	Yes	7%
Critical Design Review (CDR)	Yes	9%
OCT Interoperability Test (OIT) Success	Yes	2%
IBS-L Mission Payload Compatibility Test (ICT) Success	Yes	2%
UHF TACSATCOM Mission Payload Compatibility Test (UTCT) Success	Yes	2%
S-band TACSATCOM Mission Payload Compatibility (STCT) Test Success	Yes	2%
S-Band and Ka-Band Compatibility Test (SKCT) Success	Yes	2%
Networking and Encryption Interoperability Test (NEIT) Success	Yes	3%
System Test Readiness Review (STRR)	Yes	4%
Pre-Ship Review (PSR)	Yes	4%
Functional Acceptance Test Readiness Review (FATRR) – First Plane	Yes	4%
Functional Acceptance Test Readiness Review (FATRR) – Second Plane	Yes	4%
Functional Acceptance Review (FAR) – First Plane	Yes	5%
Functional Acceptance Review (FAR) – Second Plane	Yes	5%

The Offeror's payment plan should adhere to the payment schedule provided in Table 3-4.

Table 3-4 – Payment Schedule

Government Fiscal Year	Cumulative % of Total Price, excluding O&S
FY25	8%
FY26	35%
FY27	60%
FY28	80%
FY29	100%

The Offeror's payment plan should also avoid milestone payments scheduled to occur between 1 October and 15 November of every calendar year.

4.4.2.4.1 Delivery Incentive

In addition to the negotiated price and milestone payment plan, the Government offers a delivery incentive for the successful, on-time delivery of the entire bidding unit. This incentive fee is only offered in full based on successful completion of the following terms and conditions:

- Delivery of the 20 SVs, integrated to the LV adapter, to the designated launch processing facility by 1 August 2028
- All SVs, and associated ground hardware and software, are fully compliant with the system technical requirements as specified in Attachment 3 – Technical Requirements Document
- Successful completion of the Functional Acceptance Review (FAR) for all 20 delivered SVs and ground systems

Failure to meet any of the terms and conditions of the incentive fee may result in forfeiture of the entire fee, however, the Performer is still eligible to complete agreement performance, collecting all milestones payments, as appropriate. The delivery incentive is fifteen (15) million dollars (\$15,000,000.00), payable upon successful completion of all terms and conditions.

The Government will not accept differing or reduced proposed terms or conditions as part of this solicitation, during agreement negotiations with successful Offerors, or during agreement performance with awardees. This incentive is **excluded from the proposal value**.

4.4.2.5 On Orbit Performance Penalty Payment

The Offeror shall propose a maximum liability of the agreement value at risk for potential on-orbit performance using the Table 3-2 below. The Maximum Liability Value for each period shall be 1/10th of the Maximum Liability proposed and shall be included in the Attachment 2 – Schedule of Milestone Payments, On Orbit Performance Section.

Table 3-2 – On Orbit Performance Penalty Payment

On Orbit Performance Period	Estimated Assessment Date	Maximum Liability \$ Value	Performance Assessment Score Percentage	Performance Penalty Payback Value
1	TBD			
2	TBD			
3	TBD			
4	TBD			
5	TBD			
6	TBD			
7	TBD			
8	TBD			
9	TBD			
10	TBD			
Total Potential Performance Liability				

4.4.3 Proposal Pricing Details

The Offeror shall provide the total price and elements of price broken down by WBS, Government Fiscal Year, Major Teammates, and milestone payment schedule, as follows:

1. Price by work breakdown structure (WBS) task and Government Fiscal Year (GFY) using the template in Table 3-6. The price shall be provided for items down to Level 3 of the WBS for each GFY.

Table 3-6 – Template for Pricing of Tasks by WBS and Government Fiscal Year

WBS #	WBS Element	GFY 25	GFY 26	GFY 27	GFY 28	GFY 29	GFY 30	GFY 31	GFY 32	GFY 33	Total

2. A summary of projected funding requirements by month and year per Government Fiscal Year to enable fiscal and milestone payment planning.
3. Price and percentage of price by the Prime and Major Teammates, where the minimum set of Major Teammates is defined in Section 3.
4. Milestone payment schedule based on the template provided as Attachment 2 – Schedule of Milestones and Payments.

The Government expects the total price as presented in each of the above breakdowns to be the same value. The price information should also be provided in a separate Microsoft Excel (.xlsx) file.

The Offeror shall identify pricing assumptions which may require incorporation into the resulting award instrument (e.g., use of Government Furnished Property/Facilities/Information, access to Government Subject Matter Experts, etc.) for all members of the team.

For the purposes of pricing, the Offeror shall assume that the Test and Checkout Center is located near one of the SDA Operations Centers, which are located on Grand Forks AFB, ND and Redstone Arsenal, Huntsville, AL.

4.4.3.1 Additional Pricing Data

The Offeror shall provide a summary of pricing in accordance with the template below in Microsoft Excel format (.xlsx). The value specified as the Total Price in the last row of Table 3-6 and 3-7 shall represent the total price of the proposal and will be evaluated by the Government for **price reasonableness during source selection**.

Table 3-7 –Pricing Table

Item	Description	Price
1	20 T3TL-Upsilon SVs	Price of twenty (20) T3TL-Upsilon SVs
2	T3TL-Upsilon SEIT/PM	Price of system-level SEIT/PM for twenty (20) SVs
3	T3TL-Upsilon Launch and LV Integration	Price of SV to LV integration for the twenty (20) T3TL-Upsilon SVs
4	Performer Ground Elements	Price of NOVA Ground System and FlatSat
5	Operations and Sustainment (O&S) 2-Year Base Period	Price of O&S for 2-year base period; includes LEOPS
6	O&S Option Year 1	Price of O&S for first option year
7	O&S Option Year 2	Price of O&S for second option

		year
8	O&S Option Year 3	Price of O&S for third option year
9	Delivery Incentive	\$15,000,000.00
10	Total Baseline Price	Total Price: Sum of proposed summary prices

4.4.4 Government-Furnished Equipment, Information, and Property

The Offeror shall compile a consolidated list of **all** Government-furnished equipment, information, or property being requested by themselves and **all** team members for ease of Government evaluation. This list shall clearly delineate the item, need date, and the risk associated with non- or late delivery on overall program schedule, cost, or technical performance. The Offeror may use the list provided by the Government in Attachment 5 – Government Furnished Items (GFX) List as the starting point for their proposed list of Government-furnished content.

4.4.5 Major Teammate Proposals

Offerors are responsible for compiling and providing all Major Teammate proposals for the primary agreements officer (PAO), where a Major Teammate is defined in the introduction to this volume. Teammate proposals should include all Intercompany Work Transfer Agreements (IWTAs) or similar arrangements. All proprietary teammate proposal documentation shall be prepared at the same level of detail as that required of the Prime. Any such documentation that cannot be provided with the proposed Prime performer's proposal shall be provided to the Government either by the Prime performer or by the teammate organization via DoD SAFE, following the instructions in Section 3.2.1.

Teammate proposals will only be used to determine the price reasonableness of Offerors' proposals. No information included in any teammate proposal will be evaluated for the purpose of satisfying the requirements as stated in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document or meeting the instructions associated with Volumes I, II, III, and V of this solicitation.

4.5 Volume V – Past Performance

The page limit for Volume V depends on the number of Past Performance citations as stated below. Any pages in excess of this limit, other than those noted below as specifically excluded from the overall page limit, will be removed from the Offeror's proposal and not evaluated. Classified Past Performance citations may be submitted as attachments to Volume V within the specified page limits.

4.5.1 Executive Summary

The Offeror should introduce the current and Past Performance citations, clearly stating why they have chosen the efforts and how they best prove their ability to perform the proposed work scope. Each citation should provide specific, measurable validation of the Offeror's rationale and document successful current and Past Performance directly relevant to the current solicitation. The Government expects direct correlation between the current and Past Performance citations and the résumés submitted in Volume I – Governance. Include any classified personnel résumé experience

specifically to identify key personnel and their role in the classified effort.

4.5.2 Recent and Relevant Past Performance

The Offeror should submit at least (1) and may submit up to four (4) separate efforts by the Prime and at least one (1) and up to two (2) separate efforts for each Major Teammate. These submissions should clearly demonstrate the ability of the Offeror and its teammates to satisfy the totality of the T3TL-Upsilon requirements. The minimum set of Major Teammates is defined in Section 3. All submissions shall have a period of performance within five (5) years preceding the proposal due date. Significantly more relevance will be assigned to efforts with technical accomplishment during the five (5)-year timeframe as opposed to only operations or support activities.

For each submitted citation, the Offeror should identify:

- Customer (Name, phone, email, affiliation)
- Scope to include number of SVs / payloads / systems / subsystems / components previously delivered or to be delivered
- Place of performance, period of performance, agreement type
- Bid cost/price, currently invoiced cost/price, final price / estimate at completion
- Specific rationale for effort selection and detailed justification of relevance
- Schedule, technical and/or programmatic issues encountered, mitigations identified and undertaken, and positive or negative results seen as a result of mitigation efforts

The Offeror is highly encouraged to use key graphics to convey important elements of their current and Past Performance. High-level graphical descriptions of the operational concept (e.g., DoDAF's OV-1) depicting numbers and locations of SVs, Gantt charts showing key program milestones planned and achieved, Earned Value charts (if applicable) showing estimates and current status, etc. are all *substantially* more valuable to the Government for relevance evaluation purposes than detailed written explanations. These artifacts should be taken directly from the efforts cited and be suitable for use by the Government in conjunction with interviews which may be conducted with the customers named in each citation.

4.5.2.1 Citations

Project citations shall not exceed four (4) pages each and should stand alone by project.

For each Past Performance citation, the Offeror shall provide the Attached Past Performance Questionnaire (Attachment XX) to the Government or industry contracting officer or program manager to complete and submit directly to the SDA in accordance with the questionnaire instructions. The Offeror shall provide a list of names and point of contact for the PPQs in the proposal. (Note PPQs completed by the Government and the PPQ POC list are not included in the page count).

Offerors are hereby notified that SDA intends to survey Contractor Performance Assessment Reports (CPARs) and to conduct interviews with customers identified in Volume V. Information obtained via either of these methods will be included in Offeror evaluations per the criteria in Section 3.3.

4.6 Volume VI – Draft Other Transaction (OT) Agreement

The Offeror shall provide a proposed OT agreement, including all of its attachments, using the

attached draft OT template included with this solicitation in MS Word format. Any proposed changes to the draft OT template by the Offeror shall be indicated with the Track Changes function and rationale provided for the proposed change.

5 Submission Instructions

SDA requires that Offerors submit only electronic proposals. Unless otherwise stated in these instructions, all Volumes along with any appendices and/or attachments shall be submitted only in .pdf format and must be submitted independently by file as SDA will not accept .zip files. Offerors shall submit their responses via DoD Secure Access File Exchange (SAFE): <https://safe.apps.mil>. Offerors shall submit one (1) original copy of each Volume and appendices and/or attachments **NO LATER THAN 1200 EDT 28 April 2025**. Offerors shall submit each proposal via DoD SAFE and all files should be included in a single DoD SAFE submission. Each file submitted must be clearly labeled with the SDA solicitation number, Prime performer organization name, Prime performer proposal title (short title recommended), and Volume number and appendix or attachment number (as appropriate). The size limit for each SAFE file transmission is 8192 MB.

SAFE Drop-Off Requests: Offerors shall request a SAFE Drop-off Request, via an email to ussf.pentagon.sda.mesg.sda-sn-25-0007-mbx@mail.mil **NO LATER THAN 1200 EDT 22 April 2025**. The Offeror shall request a SAFE Drop-off Request regardless of whether or not they have access to DoD SAFE already. The subject line shall be: T3TL $\cup$ Program Solicitation Submission SAFE Request (m of n) and the Offeror's organization name, where m is the request number and n is the total number of requests that will be made by the Offeror. If the Offeror's submission will exceed the DoD size limit, then they shall make the required number of requests to complete their submission. Within the body of the email, the Offeror shall clearly state the email address to which the Drop-off Request will be sent. The Offeror is free to request read receipts for each Drop-off request email sent. The Offeror will receive an email with their SAFE Drop-Off Request **NO LATER THAN 1200 EDT 24 April 2025**. Note that this email will not come from the email address listed above.

Within the DoD SAFE submission form, the Offeror shall check both the "Encrypt every file" check box and the "Send me an email when each recipient picks up the files" check box. The password shall be sent under separate email to the email address above with the subject line: T3TL $\cup$ Program Solicitation Submission SAFE Request (m of n) Password and the Offeror's organization name. The "Short note to the Recipients" field shall clearly identify the Offeror's organization name, the proposal name, and the number of the submission.

Proposals received after 1200 EDT 28 April 2025 will be considered late and not evaluated. Offerors whose proposals are considered late will be so informed by the SDA Agreements Officer (AO) and their proposals will be deleted without any files being opened or read. Classified appendices and/or attachments may be submitted if required. Should a classified submission be required, the Offeror must contact SDA via the email address above for submission instructions.

5.1 Submission of Major Teammate Proposals

Major Teammate proposal documentation that cannot be submitted directly by the Offeror shall be submitted by the teammate via DoD SAFE in accordance with the instructions in Section 3.2. In

addition to the details specified in that section, the email request for a SAFE Drop-off request shall include the teammate organization name in the subject line of each email. Each file submitted must be clearly labeled with the SDA solicitation number, Prime performer organization name, Prime performer proposal title (short title recommended), teammate organization name, and Volume number and appendix or attachment number (as appropriate). Teammates submitting separate documentation via DoD SAFE must send an email to the address in the Submission Instructions notifying SDA of this submission method **NO LATER THAN 1200 EDT 22 April 2025**.

6 Proposal Evaluations

The Government will evaluate offers using a competitive best value trade-off process among price and the non-price Factors listed herein. The Government may elect to award to other than the lowest priced Offerors, or other than the highest technically rated Offerors. The Offeror's initial proposal should contain the Offeror's best terms regarding price and technical capability.

In making its award decision, the Government will use the evaluation Factors described in the following sections. The Offeror should not assume the Government will give credit for any capability or knowledge unless it is specified in the proposal. Proposals must conform to the solicitation's requirements to be eligible for award. The Government may reject any or all proposals, if such action is in the best interest of the Government and waive informalities and minor irregularities in proposals received.

While the Government strives for maximum objectivity in its evaluations, the source selection process, by its nature, is subjective, and, therefore, professional judgment is implicit throughout the entire process.

During the best value tradeoff process, the Government may also consider technical and programmatic risk across its portfolio of programs and contracts for both SV Prime performers and key mission payload providers, when making the selection determination.

6.1 Evaluation Methodology

The following Factors will be used to evaluate each proposal. The basis for evaluation for all Factors will be the Offeror's full proposal. Factors 1–4 are listed in descending order of importance. When combined, Factors 1-4 are approximately equal to Factor 5, Price. After completion of the initial evaluation, the Government will assign an Overall Proposal Confidence rating.

The Government will evaluate information that the Offeror(s) provide in response to Section 3.1 Proposal Preparation Instructions. Proposals must conform to this solicitation. Any submitted proposal that does not adhere to the requirements in this solicitation may be eliminated from the competition and become ineligible for award.

Offeror(s) are required to meet or exceed all solicitation requirements and technical requirements or to explicitly identify anticipated waivers, in addition to those identified as Factors. Failure to comply with the terms and conditions of the solicitation may result in the Offeror(s) being ineligible for award. The Government reserves the right to cross-reference between volumes and sections of volumes to support proposal evaluation.

6.1.1 Factor 1 – Schedule

For Factor 1 – Schedule, the Government will assess the Offeror's proposed schedule and their demonstrated ability to satisfy all requirements contained in the Attachment 1 – Statement of Work

and Attachment 3 – Technical Requirements Document, in a timely fashion.

The basis of the assessment will be the Offeror's schedule and schedule margin as detailed in *Volume III – Integrated Master Schedule* of their proposal, the Master Equipment List (MEL), the Manufacturability section of proposal *Volume II – Technical*, and the submitted video of the manufacturing processes. The Government may also consider any schedule-related information provided within any of the other proposal volumes. Identified strengths and weaknesses in technical or management areas that affect schedule reasonableness or realism or contribute to schedule risk will also be considered in this assessment.

The Government will evaluate the schedule reasonableness and realism based on:

- The Offeror's demonstrated understanding of tasks and activities required to accomplish the work
- The estimated duration of tasks and activities based on technical and manufacturing/production maturity
- The sequencing of these tasks and activities
- The identification of any dependencies and other schedule relationships (e.g., subcontractor/supplier delivery dates and delivery dates of Government-furnished material)
- The critical path identification and analysis
- The identified schedule risks and risk response strategies
- The proposed milestones and performer and/or Government decision gates
- The schedule performance criteria used to manage progress against the integrated master baseline schedule
- Evaluated strengths and weaknesses in technical or management areas that affect schedule reasonableness or realism or contribute to schedule risk

For those Offerors currently providing hardware, software or other solutions to SDA based on existing efforts, key elements of the overall schedule risk evaluation will be based on a portfolio level assessment of existing and proposed work.

6.1.2 Factor 2 – Technical Solution

For Factor 2 – Technical Solution, the Government will assess the Offeror's demonstration that their technical solution will meet the system requirements in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document. Evaluation credit may be given for solutions exceeding mandatory thresholds. The technical solution includes but is not limited to, the following:

- The satellite bus,
- Mission payloads and their required interfaces,
- Telemetry, Tracking, and Command capabilities,
- Satellite system and subsystem integration and testing,
- External system interoperability,
- All Offeror-provided ground elements and their interfaces to external ground systems,
- Multi-vehicle integrated stack and launch vehicle integration,
- Launch and Early Operations (LEOPS) and,
- Operations and sustainment.

The basis of the assessment will be the Offeror's proposed technical solution and its mission utility, as provided in proposal *Volume II – Technical* and its appendices and attachments.

The Government will evaluate the technical solution based on the fidelity of technical details provided on the technical design, evidence-based Technical Readiness Level (TRL) and Manufacturing Readiness Level (MRL) assessments for subsystem or components, analysis or engineering products demonstrating the proposed design can meet Statement of Work and Technical Requirements Document specifications, and results of any risk reduction efforts to date or detailed plans for risk mitigation during program execution.

6.1.3 Factor 3 – Past Performance

For Factor 3 – Past Performance, the Government will assess the Offeror's probability of meeting the solicitation requirements based on the Offeror's demonstrated record of work performance.

The Government will evaluate Factor 3 based on information provided within proposal *Volume V – Past Performance*. The Government may consider Past Performance information not provided by the Offeror(s) including, but not limited to:

- Contractor Performance Assessment Reporting System (CPARS)
- Publicly available information
- Federal Awardee Performance and Integrity Information System (FAPIIS)
- Electronic Subcontract Reporting System (eSRS)
- Interviews with Customer Program Managers, Contracting Officers, Fee Determining Officials or other personnel with managerial or oversight responsibilities related to the cited effort
- Past Performance questionnaires completed by customers from whom the Offeror has received previous awards

The Government will evaluate the Offeror's **recent** (performance was within five (5) calendar years of the proposal due date) and **relevant** record of Past Performance information obtained to determine whether Past Performance efforts relate to the scope of activities described in the Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document.

The Government will review the Past Performance information collected and determine the **quality** of the Offeror's performance, general trends, and the usefulness of the information and incorporate these into the performance confidence assessment. **More relevant Past Performance will have more influence on the performance confidence assessment than Past Performance of lesser relevance.** In evaluating Past Performance, the Government will consider the Offeror's previously demonstrated:

- Past Performance, history, and experience.
- Ability to accomplish requirements to receive milestone or performance-based payments. This includes activities such as interim deliverables, delivery of CDRLs or other technical and business reports, and completion of customer direction through task assignments or technical directions.
- Ability to meet technical requirements and performance standards for previous work.
- Ability to meet delivery or performance dates.
- Approach in determining probable root cause for less than fully successful missions and resultant actions taken to improve reliability.

- Safety and mission assurance performance record, including any safety mishaps and associated resolution.

The Government will consider recent or on-going SV development contracts or agreements with SDA, such as Tranche 0 Transport Layer (T0TL), Tranche 1 Transport Layer (T1TL), and Tranche 1 Demonstration and Experimentation System (T1DES), Tranche 2 Transport Layer (T2TL) as very relevant and, therefore, Performer performance on those efforts will be heavily considered and weighted in the evaluation of Factor 3 even if not specifically submitted by the Offeror.

6.1.4 Factor 4 – Technical Management and Processes

For Factor 4 – Technical Management and Processes, the Government will assess the Offeror's overall management capability and demonstrated ability to meet the work performance requirements in Attachment 1 – Statement of Work and Attachment 3 – Technical Requirements Document.

The basis of the assessment will be the Offeror's proposed technical management and processes approach as provided in proposal *Volume I – Governance* and its appendices and attachments, the Manufacturability section of proposal *Volume II – Technical*, and the submitted video of the manufacturing processes. The Government may also consider any management-related information provided within any of the other proposal volumes.

The Government will evaluate the Technical Management approach based on:

- Demonstrated efficient and effective management methodologies, processes, and procedures to achieve Statement of Work and Technical Requirements Document specifications, including flow-down to subcontractors
- Demonstrated capabilities of key personnel who will enact the plan
- Demonstrated ability to provide Government insight into progress, risks and issues
- Demonstrated ability to execute the Statement of Work and Technical Requirements Document without conflicting interests or priorities.

The Government will evaluate the adequacy and sufficiency of the manufacturing processes and capability of the Prime and Major Teammates. The Government will evaluate the extent of participation of small businesses, nontraditional defense contractors, and/or nonprofit research institutions. Finally, the Government will assess the implications of the Offeror's asserted technical data rights and how they could impact the Government's ability to affordably operate and sustain T3TLv over the life of the constellation.

6.1.5 Factor 5 – Overall On-Orbit Performance Confidence Rating

The Government will assign an Overall On-Orbit Performance Confidence Rating to the entire proposal based on the Offeror's proposed maximum liability for the On Orbit Penalty Payback.

Table 3-8 Overall Proposal Confidence Rating

Confidence Rating	Description
High Confidence	The Offeror proposed at least 10% of the total proposal value (including options) as Maximum Liability for the On Orbit Penalty Payback.

Substantial Confidence	The Offeror proposed at least 7.50% but less than 10% of the total proposal value (including options) as Maximum Liability for the On Orbit Penalty Payback.
Satisfactory Confidence	The Offeror proposed at least 5.00% but less than 7.50% of the total proposal value (including options) as Maximum Liability for the On Orbit Penalty Payback.
Low Confidence	The Offeror proposed less than 5.00% of the total proposal value (including options) as Maximum Liability for the On Orbit Penalty Payback.

6.1.6 Factor 6 – Price

The Government will evaluate the Offeror's pricing for reasonableness, overall price, and consistency with the profile identified in Table 3-4 as part of a best value analysis as described in Section 3.3. Elements of the proposed price that may be used in the evaluation include comparison of proposed prices received in response to the solicitation; comparison of proposed prices with independent Government cost estimates; and analysis of data other than certified cost or pricing data provided by the Offeror. The Government may use data external to the Offeror's proposal such as, but not limited to, field pricing reports, industry information, Government estimates, same or similar DoD contracts, and commercial data when evaluating price reasonableness.

The Government will calculate the total price for the evaluation by adding the total price for all options to the total price for the basic requirement. Evaluation of options will not obligate the Government to exercise the option(s).

6.2 Use of Non-Government Advisors

The Offeror is advised that technical and price data submitted to the Government in response to this solicitation may be released to non-Government advisors that have signed non-disclosure agreements for review and analysis. These non-Government advisors are employed by the following entities:

- AirIn Technologies, Inc.
- Alpha 2, Inc.
- Altus LLC
- Amentum Services, Inc. (formerly CENTRA)
- Bryce Space & Technology, LLC
- Custom Analysis LLC
- Emolbi, LLC
- Full Rev LLC
- GoLion Company
- MMAX, Inc.
- MTSI
- PL Consulting, Inc.
- QinetiQ
- Qualis Corporation
- Sigmatech, Inc.
- Space Systems Integration, LLC

- Strategic Analysis, Inc.
- Strategic Engineering Solutions, LLC
- TEC Solutions, LLC
- The Stacia Group, LLC
- The Aerospace Corporation
- The MITRE Corporation

All non-Government advisors performing in this role are expressly prohibited from performing SDA-sponsored technical research and are bound by appropriate nondisclosure agreements. Any objection to release of the Offeror's proposal information to non-government advisors shall be provided in writing to the Contracting Officer within ten (10) days of the date of issuance (14 March 2025) and shall include a detailed statement of the basis for the objection. The detailed statement shall identify the specific portions of the proposal for which the Offeror objects to disclosure to non-Government advisors.

Advisors from the above entities will continue to support SDA with program management activities throughout the duration of the program. These activities will require ongoing access to technical/price data generated under the agreements as well as the ability to participate in program reviews or technical interchange meetings. Any advisors will be bound by appropriate nondisclosure agreements with SDA. Performers may request copies of the nondisclosure agreements for individual advisors post-award. SDA advisors will not enter into separate NDAs with the performers.

6.3 Number of and Scope of Agreements to be Awarded

The Government intends to select up to two (2) Offers, resulting in the total procurement of four (4) operational planes of T3TL-Upsilon SVs. The Government reserves the right to make fewer than two (2) awards, more than two (2) awards, partial awards, or make no awards at all.

6.4 Negotiations and Award

As soon as the evaluation of proposals is complete, the Offerors will be notified that (1) the proposal has been selected for funding, subject to OT Agreement negotiations, or (2) the proposal has not been selected for funding. If the Government cannot come to terms with an Offeror selected for funding, it reserves the right to exclude that Offeror and to open negotiations with other Offerors that were not initially selected. Alternatively, the Government may defer a funding decision and first establish a competitive range. If a competitive range is established the Government will conduct negotiations with each offeror included in the competitive range and request final proposal updates. The final proposal updates will be reevaluated in accordance with the criteria set forth in this solicitation. All notifications will be sent via Electronic Mail to the Technical and Administrative POCs identified on the proposal coversheet. The Agreements Officer will send successful/unsuccessful notifications via Electronic Mail to the Technical and Administrative POCs identified on the proposal coversheet.

The final step of the process will be for the selected performer(s) to negotiate an OT agreement with SDA. Work will commence after the parties execute the program agreement under the OT.

6.5 Other Administrative Information

Assuming the receipt of one or more suitable proposals, the Government reserves the right to select

for negotiation all, one, or none of the proposals received in response to this solicitation.

Attachments

XX. Past Performance Questionnaire (PPQ)

XX. Résumé Template

XX. Personnel & Facilities Template

XX. Personnel Matrix

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